

Jason Tsao

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EDUCATION

San Jose State University

San Jose, CA

Bachelor of Science in Computer Engineering, Minor in Japanese (GPA 3.70)

Aug. 2024 – Dec. 2027

- **Relevant Coursework:** Data Structures and Algorithms; Object-Oriented Concepts & Methodology; Site Reliability Engineering for Cloud

TECHNICAL SKILLS

Languages: C/C++, Assembly, JavaScript, TypeScript, HTML/CSS, Python, Bash/Zsh, PowerShell

Frameworks & Libraries: React.js, Vite, Node.js, Express, FastAPI, Prisma, SQLAlchemy, SQLite, OpenCV, OpenAI Whisper, GPT-4V, MCP

Infrastructure & DevOps: Docker, Proxmox, pfSense, Nginx, Prometheus, Grafana, Ollama, WebSockets, TCP/IP, NAT, DHCP, DNS, VLAN, VPN, SSL/TLS, Linux, Git, GitHub Actions, CI/CD

Security & Endpoint Management: Qualys, Sophos, Jamf, Intune, Active Directory, RBAC, Windows Server

EXPERIENCE

IT Systems Administrator

May 2026 – Present

Student Union, Inc. of SJSU

San Jose, CA

- Managed Proxmox VE, VMware ESX, Veeams backups, UPS and high availability for hardware service continuity.
- Implemented Kubernetes for containerized workloads, enabling continual scalable deployments for 40k students.
- Developed and deployed a server-side AI agent to orchestrate MCP calls and integrate external tools and services.
- Deployed and administered Bitbucket and Plane, configuring RBAC, branching, and CI/CD workflows.
- Diagnosed connectivity issues spanning SSL certificates, TCP/IP, DHCP, DNS, VLANs, NAT rules, and firewalls.
- Configured CUPS/SNMP for real-time printing/tracking, asset management, and automated supplies ordering.

IT Technician

Feb. 2026 – May 2026

Student Union, Inc. of SJSU

San Jose, CA

- Managed Windows print server, imaging, software installation, workstation deployment, and patch management.
- Managed and secured Windows and macOS endpoints with Intune and Jamf, enforcing compliance policies.
- Identified, prioritized, and remediated CVEs using Qualys, leveraging CVSS severity to reduce vulnerabilities.
- Assisted with MDF/IDF infrastructure support, including drop ports, switches, patching, and punchdowns.

Software Engineer, Development Team

Jan. 2026 – Present

Software and Computer Engineering Society

San Jose, CA

- Built a YouTube-to-Raspberry-Pi music streaming app in TypeScript with React, Express, Prisma, and SQLite.
- Designed an egress-only WebSocket bridge so the Pi dials out to the backend, removing inbound firewall rules.
- Containerized services with Docker Compose and ran the Pi daemon as a systemd unit piping yt-dlp into mpv.

PROJECTS

Home Lab | *Docker, Ansible, Nginx, Prometheus, Grafana, Ollama, Proxmox, pfSense, OpenMediaVault, WireGuard*

- Configured pfSense as the primary router with segmented VLANs, custom firewall rules, and NAT policies.
- Deployed Proxmox as a bare-metal hypervisor to host, isolate, and snapshot multiple homelab VMs.
- Automated host configuration and SSH key distribution with idempotent, agentless Ansible playbooks.
- Deployed OpenMediaVault with a virtual-disk RAID 10 storage pool for fault-tolerant network-attached storage.
- Configured Prometheus/Grafana aggregator for real-time metrics collection, alerting, and custom dashboards.
- Secured a self-hosted Ollama LLM endpoint behind an Nginx reverse proxy with TLS, gated by WireGuard.

StudyGuard | *Python, FastAPI, OpenCV, OpenAI Whisper, SQLAlchemy*

- Won **2nd Place** at the Seed Embodied AI Hackathon for building an AI-powered focus-monitoring system.
- Built an OpenCV and GPT-4V pipeline that classifies student focus states at 10-second intervals.
- Streamed live camera feed to a FastAPI dashboard for real-time proctoring and focus visualization.
- Used a Whisper voice agent to control Reachy Mini robot gestures and update the dnsmasq allowlist.